A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.3: Data Definition and Manipulation Languages (DDL & DML)

Concept 8.3.2: DML Queries (SELECT, JOIN, GROUP BY, Aggregate Functions)

Total Questions: 23

Mark Schemes begin on Page 20

Question 1 | Source: 9618_w25_qp_11 (Q2.c.i)

2 A relational database, `WORKEXPERIENCE`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

Part of the database is shown:

```
STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate,
          Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)
```

(c) Some example data from the `PLACEMENT` table is shown:

PlacementID	StudentID	CompanyID	StartDate	EndDate	Complete
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

(i) Write a Structured Query Language (SQL) script to delete all placements that have been completed.

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Question 2 | Source: 9618_w25_qp_11 (Q2.c.ii)

2 A relational database, `WORKEXPERIENCE`, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

Part of the database is shown:

```
STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)

PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate,
          Complete)

COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)
```

(c) Some example data from the `PLACEMENT` table is shown:

PlacementID	**StudentID**	**CompanyID**	**StartDate**	**EndDate**	**Complete**
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

(ii) Write an SQL script to return the total number of placements completed by the student with ID LDEA01 at the company with ID NEAM. The total should be given an appropriate name.

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Question 3 | Source: 9618_w25_qp_12 (Q4.c)

4 A relational database, `SHIPPING`, stores data about the ships in a company and the containers that are carried on the ships.

The database has the following tables:

```
CONTAINER(ContainerID, Type, Weight, OwnerName, ShipID)

SHIP(ShipID, Type, Capacity, ShipName)
```

(c) Write an SQL script to return the number of containers stored in the database for the ship with the name Caledonia.

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Question 5 | Source: 9618_s25_qp_11 (Q5.e)

5 A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, `FURNITURE`, has the following tables:

CUSTOMER(<u>CustomerID</u>, Name, Phone)

CUSTOMER_CARD_DATA(<u>CardID</u>, CustomerID, CardType, CardNumber, EndDate)

ORDER(<u>OrderID</u>, CustomerID, TotalCost, Paid, OrderDate, Complete)

ORDER_ITEM(<u>OrderItemID</u>, OrderID, Type, Height, Width, Depth, Details)

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

(e) Write an Structured Query Language (SQL) script to output the customer ID, the customer's name and the total cost of the customer's orders that have **not** been paid.

The output of the total cost must have an appropriate title.

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Question 6 | Source: 9618_s25_qp_12 (Q5.d.i)

5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

`USER(Username, Password, DateOfBirth)`

`CHARACTER(CharacterName, CharacterID, Username, Level, Money)`

`ITEM(ItemName, MinimumLevel, Cost)`

`CHARACTER_ITEM(CharacterID, ItemName)`

(d) **(i)** Write a Structured Query Language (SQL) script to count the number of items purchased by the user with the username "`KAT123`".

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Question 7 | Source: 9618_s25_qp_12 (Q5.d.ii)

5 An online game has a database that stores data about users and the characters each user creates in the game. Each user can create multiple characters and purchase multiple items for each character.

The normalised database has the following design:

`USER(Username, Password, DateOfBirth)`

`CHARACTER(CharacterName, CharacterID, Username, Level, Money)`

`ITEM(ItemName, MinimumLevel, Cost)`

`CHARACTER_ITEM(CharacterID, ItemName)`

(ii) The following changes need to be made to the character with the ID `"0002"`:

- level changed to `3`
- money changed to `10000.00`

Write an SQL script to change the character's data.

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Question 8 | Source: 9618_s25_qp_13 (Q6.c)

6 A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

`CUSTOMER(CustomerID, CustomerName, Email)`

`ORDER(OrderID, CustomerID, Date, Collected)`

`ORDER_PRODUCT(OrderID, ProductID, Quantity)`

`PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)`

`SUPPLIER(SupplierID, SupplierName, SupplierEmail)`

The primary keys are underlined.

(c) The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

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Question 9 | Source: 9618_w24_qp_11 (Q2.b.iii)

2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

PART(<u>PartID</u>, Description, Price, SupplierID)

CUSTOMER(<u>CustomerID</u>, FirstName, LastName, ContactNumber)

REPAIR(<u>RepairNumber</u>, StartDate, EndDate, CustomerID, Device)

REPAIR_PART(<u>PartID</u>, <u>RepairNumber</u>, Quantity)

(iii) Suppliers send invoices to the company for the parts that are used. A new table, `INVOICE`, stores the data about each invoice and whether it has been paid or not.

The design for the table `INVOICE` is shown:

`INVOICE(InvoiceID, SupplierID, AmountDue, Paid, DatePaid)`

The table shows sample data for the table `INVOICE`.

InvoiceID	**SupplierID**	**AmountDue**	**Paid**	**DatePaid**
000001	JK675	22.50	TRUE	01/01/2024
000002	WR443	358.99	FALSE	
000003	JK675	10.21	FALSE	

Write an SQL script to return the total amount due to the supplier with the ID of JK675 for all the invoices that have **not** currently been paid.

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Question 10 | Source: 9618_w24_qp_12 (Q6.b.ii)

6 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(b) The company decides which employees will work on each repair job. An employee can log into the database to access information about their repair jobs.

The database is normalised and includes these tables:

- `CUSTOMER` stores personal data about each customer
- `EMPLOYEE` stores personal data about each employee
- `LOGIN_DATA` stores the username and password for each employee
- `JOB` stores the data about each repair job
- `JOB_EMPLOYEE` stores the employees that are working on each repair job.

(ii) The database also has the table `INVOICE` that stores data about each invoice that is sent to a customer.

Example data from the table `INVOICE` is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

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Question 11 | Source: 9618_w24_qp_13 (Q4.b)

4 A company makes ice cream and sells it to shops.

The ice cream is made in batches: a large quantity of one type and flavour of ice cream that is then split into smaller quantities for sale.

The company's owner has designed a relational database, `ICECREAM`, to store data about their ice cream and customers.

Some of the tables in the database are given. The database is not normalised.

```
BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

CUSTOMER(CustomerID, CompanyName, EmailAddress, TelephoneNumber)

SALE(SaleID, BatchID, CustomerID, Quantity, Date)
```

(b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

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Question 12 | Source: 9618_s24_qp_11 (Q6.c.ii)

6 A company is developing a website that will allow users to create an account and then play a quiz every day. The data about the users and the quizzes are stored in a database.

A user must select a unique username and enter a valid email address to create an account. All users must be over the age of 16. A new quiz is given to the users every day. Each quiz is stored in its own text file.

The database stores the filename of each quiz and the date it can be played. The user gets a score for each quiz they complete, which is stored in the database. The scores are used to give each user a rating, for example Gold.

(c) The company has another database, `FARMING`, for a different game.

The database `FARMING` has a table named `EVENT` which is shown with some sample data.

PlayerID	**EventID**	**Category**	**Points**
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

(ii) Write an SQL script to return the number of events that each player has completed.

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Question 13 | Source: 9618_s24_qp_13 (Q4.c)

4 A theatre wants to use a database to store data about the shows that are scheduled, their customers and the seats that the customers have booked.

In the theatre:

- Each show can take place on multiple dates.
- Some dates can have more than one performance.
- There are multiple rows of seats.
- Each seat can be individually booked by its row letter and seat number, for example row E seat 2.

Part of the database design includes these tables:

```
SHOW(ShowID, Title, Duration)

SEAT(SeatID, RowLetter, SeatNumber)

PERFORMANCE(PerformanceID, ShowID, ShowDate, StartTime)
```

(c) Write an SQL script to return the number of times each show is scheduled. For example, in the sample data in part **(b)**, the show MK12 is scheduled three times.

The result needs to include the show name and a suitable field name for the number of times it is scheduled.

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Question 14 | Source: 9618_w23_qp_11 (Q3.c.ii)

3 A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

```
CUSTOMER(CustomerID, FirstName, LastName, Town)

SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)
```

(ii) Write the SQL script to return the total quantity of items that the customer with the ID of HJ231 has ordered.

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Question 15 | Source: 9618_w23_qp_13 (Q3.b.ii)

3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

(b) The company uses a database, `COURSES`, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

The database has the following structure:

COURSE_INFORMATION(<u>CourseID</u>, Description, Cost)

TUTOR(<u>TutorID</u>, TelephoneNumber, EmailAddress, TutorName)

COURSE_SCHEDULE(<u>CourseID</u>, <u>DateStarted</u>, TutorID)

(ii) Write the Structured Query Language (SQL) script to return the total number of courses that have started after 9 September 2023.

The value returned must have an appropriate field name.

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Question 16 | Source: 9618_s23_qp_11 (Q2.b.iv)

2 An organisation uses a database to store data about the types of bird that people have seen.

(b) The database, `Birds`, stores information about the types of bird and the people who have seen them.

Data about each bird seen is stored with its location and data about the person who saw the bird.

Database `Birds` has the following tables:

```
BIRD_TYPE(BirdID, Name, Size)

BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)

PERSON(PersonID, FirstName, LastName, EmailAddress)
```

(iv) The database tables are repeated here for reference:

```
BIRD_TYPE(BirdID, Name, Size)

BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)

PERSON(PersonID, FirstName, LastName, EmailAddress)
```

Complete the SQL script to return the number of birds of each size seen by the person with the ID of J_123.

```
SELECT BIRD_TYPE.Size, ........................................ (BIRD_TYPE.BirdID)

       AS NumberOfBirds

FROM BIRD_TYPE, ........................................

WHERE .................................................. = "J_123"

AND BIRD_TYPE.BirdID = ..................................................

........................................ BIRD_TYPE.Size;
```

[5]

Question 17 | Source: 9618_s23_qp_12 (Q2.c.ii)

2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

(c) The database `Lessons` has the following tables:

HORSE(<u>HorseID</u>, Name, Height, Age, HorseLevel)

STUDENT(<u>StudentID</u>, FirstName, LastName, RiderLevel, PreferredHorseID)

LESSON(<u>LessonID</u>, Date, Time, StudentID, HorseID, LessonContent)

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(ii) Write a Structured Query Language (SQL) script to return the names of all the horses that have the horse level intermediate or beginner.

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Question 18 | Source: 9618_s23_qp_12 (Q2.c.iii)

2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

(c) The database `Lessons` has the following tables:

`HORSE(`<u>`HorseID`</u>`, Name, Height, Age, HorseLevel)`

`STUDENT(`<u>`StudentID`</u>`, FirstName, LastName, RiderLevel, PreferredHorseID)`

`LESSON(`<u>`LessonID`</u>`, Date, Time, StudentID, HorseID, LessonContent)`

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders

FROM STUDENT, LESSON

WHERE StudentID = StudentID

OR Date = #09/09/2023#

AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

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[4]

Question 19 | Source: 9618_w22_qp_11 (Q4.c.ii)

4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, `PHOTOGRAPHS`, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

CUSTOMER(<u>CustomerID</u>, FirstName, LastName, Telephone)

PARTY(<u>PartyID</u>, CustomerID, PartyDate, StartTime)

PHOTO_DATA(<u>PhotoID</u>, PartyID, TimeTaken, CameraID)

CAMERA_DATA(<u>CameraID</u>, LensType, LightingType)

(c) The table shows some sample data for the table `PHOTO_DATA`.

PhotoID	PartyID	TimeTaken	CameraID
ST23-56	BD987	08:34	NIK-02
ST23-57	BD987	08:55	NIK-02
ST23-60	BC08	09:01	CAN-01
ST23-61	BC08	10:23	CAN-12
ST23-62	BC08	10:56	NIK-01

(ii) Complete the Structured Query Language (SQL) script to display the total number of photographs that have been taken using a camera with a camera ID starting with `CAN`.

```
SELECT ....................................................... (...............................................................)
FROM ..........................................................................................
WHERE CameraID LIKE ................................................................................. ;
```

[4]

Question 20 | Source: 9618_w22_qp_13 (Q2.c)

2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

```
COMPANY(TelephoneNumber, CompanyID, CompanyName)

PHOTOGRAPH(PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)

TELESCOPE(TelescopeID, CompanyID, SerialNumber)
```

(c) Complete the SQL script to return the total number of telescopes owned by the company whose ID begins with `HW`.

```
SELECT .......................................... (...........................................................)
FROM TELESCOPE
WHERE ................................................. LIKE ..............................................;
```

[4]

Question 21 | Source: 9618_w21_qp_11 (Q5.c.i)

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

```
SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)

MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)

CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)
```

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

(i) Complete the following DML statements to return the number of cars for sale in each shop.

```
SELECT COUNT(.....................................................)

FROM .....................................................

..................................................... ShopID
```

[3]

Question 22 | Source: 9618_w21_qp_11 (Q5.c.ii)

5 Javier owns many shops that sell cars. He employs several managers who are each in charge of one or more shops. He uses the relational database `CARS` to store the data about his business.

Part of the database is shown:

```
SHOP(ShopID, ManagerID, Address, Town, TelephoneNumber)

MANAGER(ManagerID, FirstName, LastName, DateOfBirth, Wage)

CAR(RegistrationNumber, Make, Model, NumberOfMiles, ShopID)
```

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

(ii) Complete the DML statement to include the following car in the table `CAR`.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

.... CAR

.... ("123AA","Tiger","Lioness",10500,"12BSTREET")

[2]

Question 23 | Source: 9618_w21_qp_12 (Q6.c.i)

6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, `PLANTSALES`, has the following structure:

PLANT(<u>PlantName</u>, QuantityInStock, Cost)

CUSTOMER(<u>CustomerID</u>, FirstName, LastName, Address, Email)

PURCHASE(<u>PurchaseID</u>, CustomerID)

PURCHASE_ITEM(<u>PurchaseID</u>, <u>PlantName</u>, Quantity)

(c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

(i) Complete the DML statements to return the total number of items purchased with the purchase ID of 3011A.

SELECT SUM(..)

FROM ....

WHERE .... = ...;

[4]

Question 24 | Source: 9618_s21_qp_12 (Q1.c.ii)

1 Raj owns houses that other people rent from him. He has a database that stores details about the people who rent houses, and the houses they rent. The database, `HOUSE_RENTALS`, has the following structure:

```
CUSTOMER(CustomerID, FirstName, LastName, DateOfBirth, Email)
HOUSE(HouseID, HouseNumber, Road, Town, Bedrooms, Bathrooms)
RENTAL(RentalID, CustomerID, HouseID, MonthlyCost, DepositPaid)
```

(c) Example data from the table `RENTAL` are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

(ii) Write a Data Manipulation Language (DML) script to return the first name and last name of all customers who have **not** paid their deposit.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q2.c.i)

2(c)(i)	**1 mark** per bullet point, **max 2 marks**
	
	• DELETE FROM and correct table
	• Correct condition
	
	Example answer:
	`DELETE FROM PLACEMENT`
	`WHERE Complete = TRUE;`
		2

Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q2.c.ii)

2(c)(ii)	**1 mark** per bullet point, **max 4 marks**
	
	• SELECT COUNT any appropriate field
	• AS and an appropriate name
	• FROM and correct table and WHERE and one correct condition
	• Two AND clauses with the other two correct conditions
	
	Example answer:
	`SELECT COUNT (CompanyID)`
	`AS TotalPlacements`
	`FROM PLACEMENT`
	`WHERE CompanyID = "NEAM"`
	`AND StudentID = "LDEA01"`
	`AND Complete = TRUE;`
		4

Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q4.c)

4(c)	**1 mark** per bullet point, **max 4** marks
	
	• SELECT <u>COUNT</u> statement
	• Using the correct tables
	• Joining the tables
	• The condition for the name
	
	Example Answer 1:
	
	`SELECT COUNT(ContainerID)`
	`FROM CONTAINER, SHIP`
	`WHERE CONTAINER.ShipID = SHIP.ShipID`
	`AND ShipName = "Caledonia";`
	
	Example Answer 2:
	
	`SELECT COUNT(ContainerID)`
	`FROM CONTAINER INNER JOIN SHIP`
	`ON CONTAINER.ShipID = SHIP.ShipID`
	`WHERE ShipName = "Caledonia";`
		4

Mark Scheme for Question 4 | Source: 9618_w25_ms_13 (Q5.c)

5(c)	**1 mark** per bullet point, **max 5** marks
	
	• SELECT and the correct attributes
	• FROM and the correct tables
	• Tables joined correctly
	• Correct condition for the rating
	• Correct ORDER BY clause
	
	Example Answer 1
	`SELECT PRODUCT.ProductID, ProductName, ComplaintDetails`
	`FROM PRODUCT, COMPLAINT`
	`WHERE PRODUCT.ProductID = COMPLAINT.ProductID`
	`AND Rating <=5`
	`ORDER BY Rating DESC;`
	
	Example Answer 2
	`SELECT PRODUCT.ProductID, ProductName, ComplaintDetails`
	`FROM PRODUCT INNER JOIN COMPLAINT`
	`ON PRODUCT.ProductID = COMPLAINT.ProductID`
	`WHERE Rating <=5`
	`ORDER BY Rating DESC;`
		5

Mark Scheme for Question 5 | Source: 9618_s25_ms_11 (Q5.e)

5(e)	**1 mark** each • Selecting the customer ID, customer name and sum of TotalCost with appropriate identifier • FROM clause with suitable join of tables (ON or WHERE) • ORDER.Paid = FALSE condition (with correct key word) • GROUP BY condition e.g. SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER INNER JOIN ORDER ON CUSTOMER.CustomerID = ORDER.CustomerID WHERE ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID; Alternative JOIN Statement SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER,ORDER WHERE CUSTOMER.CustomerID = ORDER.CustomerID AND ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID;	**4**

Mark Scheme for Question 6 | Source: 9618_s25_ms_12 (Q5.d.i)

5(d)(i)	**1 mark** for each point • Select COUNT • FROM (using correct tables) and joining • WHERE (or AND) condition for correct username Example 1: SELECT COUNT(ItemName) FROM CHARACTER_ITEM INNER JOIN CHARACTER ON CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID WHERE CHARACTER.Username = "KAT123"; Example 2: SELECT COUNT(ItemName) FROM CHARACTER_ITEM, CHARACTER WHERE CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID AND CHARACTER.Username = "KAT123";	**3**

Mark Scheme for Question 7 | Source: 9618_s25_ms_12 (Q5.d.ii)

5(d)(ii)	**1 mark** for each point • UPDATE character • SET level and money • WHERE condition e.g. UPDATE CHARACTER SET Level = 3, Money = 10000.00 WHERE CharacterID = "0002";	**3**

Mark Scheme for Question 8 | Source: 9618_s25_ms_13 (Q6.c)

6(c)	**1 mark** each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	**4**

Mark Scheme for Question 9 | Source: 9618_w24_ms_11 (Q2.b.iii)

2(b)(iii)	**1 mark** for each bullet point
	
	• Select SUM of `AmountDue`
	• From the correct table **and** one correct condition
	• Second correct condition
	
	`SELECT SUM(AmountDue)`
	`FROM INVOICE`
	`WHERE SupplierID = "JK675" AND Paid = FALSE;`
		3

Mark Scheme for Question 10 | Source: 9618_w24_ms_12 (Q6.b.ii)

6(b)(ii)	**1 mark** each
	
	• Select SUM of Amount
	• From the correct table **and** one correct condition
	• Remaining correct condition
	
	Example:
	
	`SELECT SUM(Amount)`
	`FROM INVOICE`
	`WHERE Paid = "Y"`
	`AND DateSent >= #01/01/2023# AND DateSent <=`
	`#31/12/2023#`
		3

Mark Scheme for Question 11 | Source: 9618_w24_ms_13 (Q4.b)

4(b)	**1 mark** each:
	
	• `SELECT SUM(Quantity)`
	• `FROM SALE WHERE` and one correct condition
	• `AND` with remainder correct conditions
	
	e.g.
	`SELECT SUM(Quantity)`
	`FROM SALE`
	`WHERE CustomerID = "0034E"`
	`AND Date >= #01/01/2023# AND Date <= #31/12/2023#;`
		3

Mark Scheme for Question 12 | Source: 9618_s24_ms_11 (Q6.c.ii)

6(c)(ii)	**1 mark** each:
	
	• Selecting PlayerID from EVENT
	• Counting EventID
	• Grouping by the PlayerID
	
	Example:
	`SELECT PlayerID, COUNT(EventID)`
	`FROM EVENT`
	`GROUP BY PlayerID;`
		3

Mark Scheme for Question 13 | Source: 9618_s24_ms_13 (Q4.c)

4(c)	**1 mark** each for:
	
	• Selecting `COUNT` of an attribute in `PERFORMANCE` table with suitable name
	• `FROM` clause
	• Joining tables
	• Grouping by the title and selecting the title
	
	Example 1:
	`SELECT SHOW.Title, Count(PERFORMANCE.PerformanceID)`
	`AS NumberOfShowings`
	`FROM PERFORMANCE, SHOW`
	`WHERE PERFORMANCE.ShowID = SHOW.ShowID`
	`GROUP BY SHOW.Title;`
	
	Example 2:
	`SELECT SHOW.Title, Count(PERFORMANCE.PerformanceID)`
	`AS NumberOfShowings`
	`FROM PERFORMANCE INNER JOIN SHOW`
	`ON PERFORMANCE.ShowID = SHOW.ShowID`
	`GROUP BY SHOW.Title;`
		4

Mark Scheme for Question 14 | Source: 9618_w23_ms_11 (Q3.c.ii)

3(c)(ii)	**1 mark** for each line (**max 4**)
	
	Either:
	`SELECT SUM(Quantity)`
	`FROM ORDER_ITEM, SHOP_ORDER`
	`WHERE ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo`
	`AND SHOP_ORDER.CustomerID = 'HJ231';`
	OR
	`SELECT SUM(Quantity)`
	`FROM ORDER_ITEM (INNER) JOIN SHOP_ORDER`
	`ON ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo`
	`WHERE SHOP_ORDER.CustomerID = 'HJ231';`
		4

Mark Scheme for Question 15 | Source: 9618_w23_ms_13 (Q3.b.ii)

3(b)(ii)	**1 mark** for each bullet point.
	
	• `SELECT Count(CourseID)`
	• `AS NumOfCourses`
	• `FROM COURSE_SCHEDULE`
	• `WHERE DateStarted > "09/09/23";`
	
	`SELECT Count(CourseID) AS NumOfCourses FROM`
	`COURSE_SCHEDULE`
	`WHERE DateStarted > "09/09/23";`
		4

Mark Scheme for Question 16 | Source: 9618_s23_ms_11 (Q2.b.iv)

2(b)(iv)	**1 mark** for each correctly completed space
	
	`SELECT BIRD_TYPE.Size, `**`COUNT`**`(BIRD_TYPE.BirdID) AS NumberOfBirds`
	
	`FROM BIRD_TYPE, `**`BIRD_SEEN`**
	
	`WHERE `**`BIRD_SEEN.PersonID`**` = "J_123"`
	
	`AND BIRD_TYPE.BirdID = `**`BIRD_SEEN.BirdID`**
	
	`GROUP BY`` BIRD_TYPE.Size;`
		5

Mark Scheme for Question 17 | Source: 9618_s23_ms_12 (Q2.c.ii)

2(c)(ii)	**1 mark** each
	
	• SELECT field Name
	• FROM table HORSE
	• WHERE with Intermediate / Beginner
	• OR with Beginner / Intermediate
	
	Example answer:
	`SELECT Name`
	`FROM HORSE`
	`WHERE HorseLevel = "Intermediate"`
	`OR HorseLevel = "Beginner";`
		4

Mark Scheme for Question 18 | Source: 9618_s23_ms_12 (Q2.c.iii)

2(c)(iii)	**1 mark** each • SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) • The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID • The OR should be AND // AND Date = #09/09/2023# • Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	**4**

Mark Scheme for Question 19 | Source: 9618_w22_ms_11 (Q4.c.ii)

4(c)(ii)	**1 mark** for each correctly completed empty space:
	
	• COUNT
	• PhotoID
	• PHOTO_DATA
	• 'CAN*' // 'CAN%'
	
	`SELECT` **`COUNT`**`(`**`PhotoID`**`)`
	`FROM` **`PHOTO_DATA`**
	`WHERE CameraID LIKE` **`'CAN*'`**`; // WHERE CameraID LIKE`
	`'CAN%'``;`
		4

Mark Scheme for Question 20 | Source: 9618_w22_ms_13 (Q2.c)

2(c)	**1 mark** for each correctly completed missing part:
	
	`SELECT` **`COUNT`** `(`**`TelescopeID`**`)`
	`FROM` **`TELESCOPE`**
	`WHERE` **`CompanyID`** `LIKE` **`'HW%'`**`;`
		4

Mark Scheme for Question 21 | Source: 9618_w21_ms_11 (Q5.c.i)

5(c)(i)	**1 mark** per correctly completed statement `SELECT COUNT(`**`RegistrationNumber`**`)` `FROM` **`CAR`** **`GROUP BY`** `ShopID`	**3**

Mark Scheme for Question 22 | Source: 9618_w21_ms_11 (Q5.c.ii)

5(c)(ii)	**1 mark** for each correct statement **`INSERT INTO`** `CAR` **`VALUES`** `("123AA","Tiger","Lioness",10500,"12BSTREET")`	**2**

Mark Scheme for Question 23 | Source: 9618_w21_ms_12 (Q6.c.i)

6(c)(i)	**1 mark** for each correctly completed space
	
	`SELECT SUM(`**`Quantity`**`)`
	`FROM` **`PURCHASE_ITEM`**
	`WHERE` **`PurchaseID`** `=` **`"3011A"`**`;`
		4

Mark Scheme for Question 24 | Source: 9618_s21_ms_12 (Q1.c.ii)

1(c)(ii)	**1 mark** per bullet point
	
	• Select `FirstName` and `LastName`
	• From both tables
	• Where `DepositPaid = No`
	• Joining tables (either `AND`, or `INNER JOIN`)
	
	Example script:
	`SELECT FirstName, LastName`
	`FROM CUSTOMER, RENTAL`
	`WHERE DepositPaid = No`
	`AND RENTAL.CustomerID = CUSTOMER.CustomerID;`
		4